

EE150 Signal and System

Homework 2

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1. (15pt)

Given the discrete-time signal

$$x[n] = \{1, 2, 3, 4\}, \quad n = 0, 1, 2, 3,$$

and zero elsewhere.

- (a) Use the sifting property to represent the signal $x[n]$ as the weighted sum of shifted unit impulses (the impulse expansion). (5pt)
- (b) Verify your solution in (a) at $n = 2$. (5pt)
- (c) Compute the convolution of $x[n]$ with $\delta[n - k]$, where k is a constant. (5pt)

Solution:

Exercise 2. (20pt)

Evaluate and sketch the output of the systems whose impulse responses $h[n]$ or $h(t)$ and input signals $x[n]$ or $x(t)$ are given by:

(a)(5pt) $h[n] = \left(\frac{1}{4}\right)^n u[n]$, $x[n] = u[2n]$.

(b)(5pt) $h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$, $x[n] = u[n] - u[n-2]$.

(c)(5pt) $h(t) = u(t) - u(t-3)$, $x(t) = e^{-t}u(t)$.

(d)(5pt) $h(t) = u(t) - u(t-3)$, $x(t) = \begin{cases} t+1, & 0 \leq t \leq 1 \\ 3-t, & 1 < t \leq 3 \\ 0, & \text{other} \end{cases}$

Solution:

Exercise 3. (11pt)

In the discrete-time system shown in the following figure, it is known that

$$h_1[n] = u[n]$$

$$h_2[n] = \delta[n]$$

$$h_3[n] = a\delta[n - 1]$$

If the input signal $x[n] = a^n u[n]$, find the impulse response $h[n]$ and the output signal $y[n]$ of the entire system.

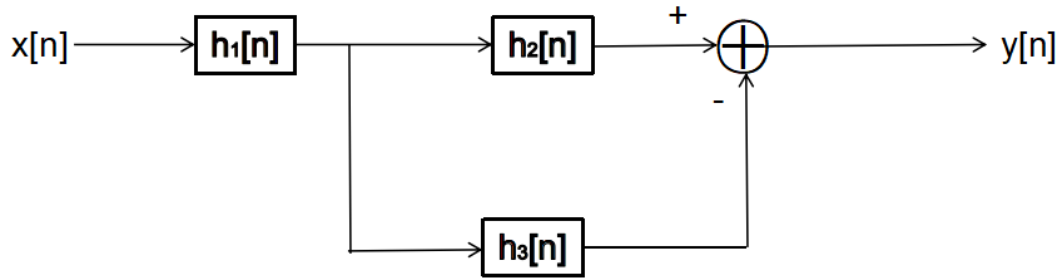


Figure 1: The block diagram of the discrete-time system.

Solution:

Exercise 4. (24pt)

Given the following LTI systems, determine the three properties of each system with simple explanation: (1) memoryless, (2) causal, (3) stable :

(a)(8pt) $h[n] = \left(\frac{1}{4}\right)^n u[n - 1]$, where $h[n]$ is the impulse response of the system.

(b)(8pt) $y[n] = \sum_{k=-\infty}^{\infty} \frac{1}{1+k^2} x[n - k]$, where $x[n]$ is the input signal and $y[n]$ is the output signal.

(c)(8pt) $y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau - 3) d\tau$, where $x(t)$ is the input signal and $y(t)$ is the output signal.

Solution:

Exercise 5. (30pt)

- (a)(15pt) Draw the block diagram representation for the causal LTI system described by the following differential equation, and determine the system output $y(t)$ when $x(t) = 3e^{-t}u(t)$ (causality of a linear system equivalent to the condition of initial rest).

$$3x(t) + \frac{dx(t)}{dt} = y(t) + 2\frac{dy(t)}{dt},$$

- (b)(15pt) Given the block diagram representations for the causal LTI systems, determine the difference equation. Then, when $x[n] = (0.5)^n u[n]$, write the homogeneous solution $y_h[n]$ and a particular solution of the same form as the input $y_p[n]$. Give the complete solution $y[n]$ (causality of a linear system equivalent to the condition of initial rest).

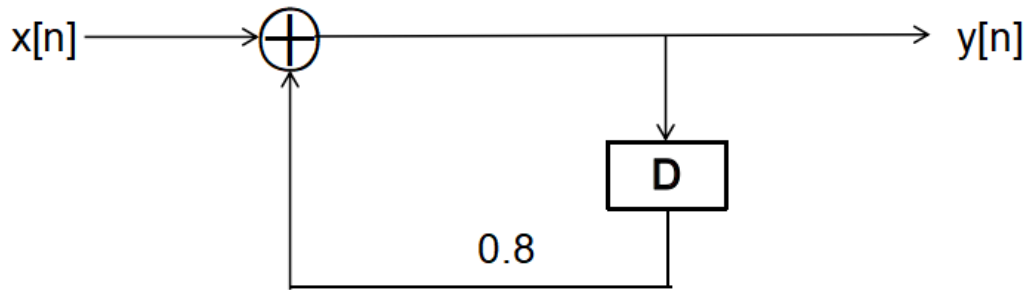


Figure 2: The block diagram of the system in 5(b).

Solution: